

Agentic Reasoning for Large Language Models

FOUNDATIONS · EVOLUTION · COLLABORATION

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UIUC · Meta · Amazon · Google DeepMind · UCSD · Yale

arXiv:2601.12538 | 2026

Roadmap

- **Background and Overview**
- Pillar 1: Foundational Agentic Reasoning
- Pillar 2: Self-Evolving Agentic Reasoning
- Pillar 3: Collective Multi-Agent Reasoning
- Applications and Benchmarks
- Open Challenges
- Discussion

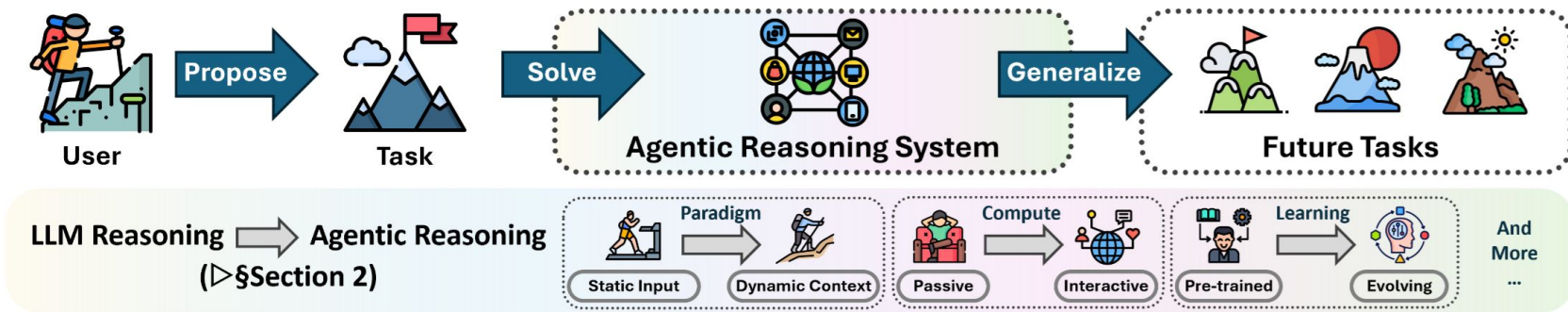
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The Challenge: LLMs in a Dynamic World

Large Language Models (LLMs) show powerful reasoning in **closed-world settings like math and code**. However, they struggle in open-ended, dynamic environments where they cannot plan, act, or learn from interaction.

The Solution: Agentic Reasoning

Agentic reasoning reframes LLMs as autonomous agents that can **plan, act, and learn through continual interaction** with their environment. This marks a **paradigm shift** from passive text generation to active, goal-oriented problem-solving.



From Passive Prediction to Agentic Interaction

Agentic reasoning transforms inference from a static, one-shot prediction into a dynamic, multi-step process involving interaction, feedback, and memory.

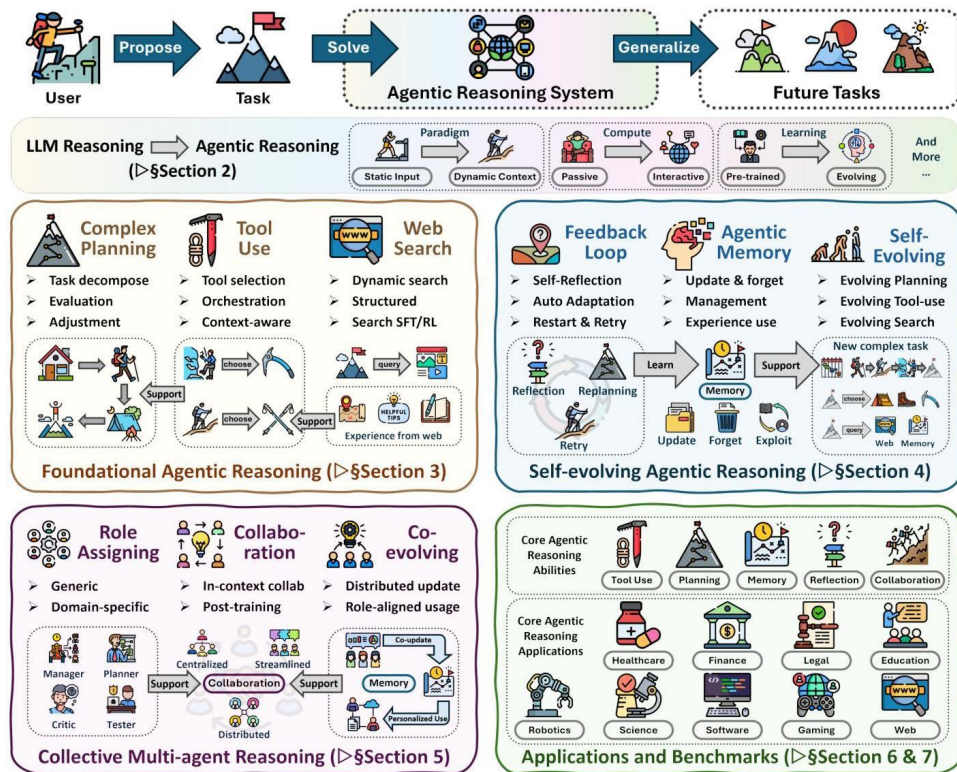
Dimension	Traditional LLM Reasoning	Agentic Reasoning
Paradigm	Passive, Static Input	Interactive, Dynamic Context
Computation	Single Pass, Internal Compute	Multi-step, With Feedback
Statefulness	Context Window, No Persistence	External Memory, State Tracking
Learning	Offline Pretraining, Fixed Knowledge	Continual Improvement, Self-Evolving
Goal Orientation	Prompt-based, Reactive	Explicit Goal, Planning

A Unified Roadmap for Agentic Reasoning

Three Complementary Dimensions

This survey organizes agentic reasoning along three key dimensions, providing a systematic map from core capabilities to complex, collaborative intelligence.

- **Foundational Reasoning:** Core single-agent skills like planning, tool use, and search.
- **Self-Evolving Reasoning:** How agents improve through feedback, memory, and adaptation.
- **Collective Multi-agent Reasoning:** Scaling intelligence to collaborative scenarios with multiple agents.



The Three Pillars of Agentic Reasoning



01

Foundational Capabilities

Core single-agent skills that form the operational backbone:

- **Planning** to decompose goals
- **Tool Use** to invoke APIs
- **Search** to navigate info



02

Self-Evolving Reasoning

Mechanisms for continual improvement without full retraining:

- **Feedback Loops** (reflection)
- **Memory Management**
- **Continual Adaptation**



03

Collective Multi-Agent

Frameworks for collaborative problem-solving:

- **Coordination** protocols
- **Role Specialization**
- **Shared Memory** systems

"Agentic reasoning positions reasoning as the central mechanism of intelligent agents, spanning foundational capabilities, self-evolving adaptation, and collective coordination."

Formalizing Agentic Reasoning: A Latent-Space View

We formalize the environment as a **Markov Decision Process**:

- **O**: observation space (search/query/tool outputs)
- **A**: external action space (tool calls, search, plan, reflect, memory retrieve, etc.)
- **Z**: reasoning trace space (reasoning thoughts)
- **M**: internal memory or context state

At time t , the agent conditions on history or integrate with the memory state:

$$h_t = (o_{\leq t}, z_{< t}, a_{< t})$$

Policy Decomposition

Agent behavior emerges from the synergy between reasoning and acting:

$$\pi_{\theta}(z_t, a_t \mid h_t) = \underbrace{\pi_{\text{reason}}(z_t \mid h_t)}_{\text{Internal Thought}} \cdot \underbrace{\pi_{\text{exec}}(a_t \mid h_t, z_t)}_{\text{External Action}}.$$

- **Internal Thought**: reasoning in latent space Z
- **External Action**: execution in environment with action space A

Agentic Reasoning Mechanisms: In-Context v.s Post-Training

In-Context Reasoning (Non-parametric Adaptation)

Improves reasoning **without updating model parameters**.

- Chain-of-Thought prompting
- Reasoning decomposition
- Test-time tool use and retrieval
- Search over reasoning trajectories

State update happens only in **context or memory**:

$$m_t = f(m_{t-1}, o_t, z_t)$$

This enables **test-time adaptation** to new tasks.

Post-Training Reasoning (Parametric Adaptation)

Improves reasoning **through parameter updates**.

Typical approaches:

- **Supervised Fine-Tuning (SFT)**
 - a. training on reasoning traces
 - b. training on acting trajectories
- **Agentic RL**
 - a. learning acting strategies
 - b. optimizing long-horizon decisions

Post-training converts reasoning behaviors into **stable parametric capabilities**.

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Pillar 1: Foundational Agentic Reasoning

This layer focuses on how an individual agent translates reasoning into structured action through three core components. The agent becomes an autonomous problem solver, not just a text generator.

Planning

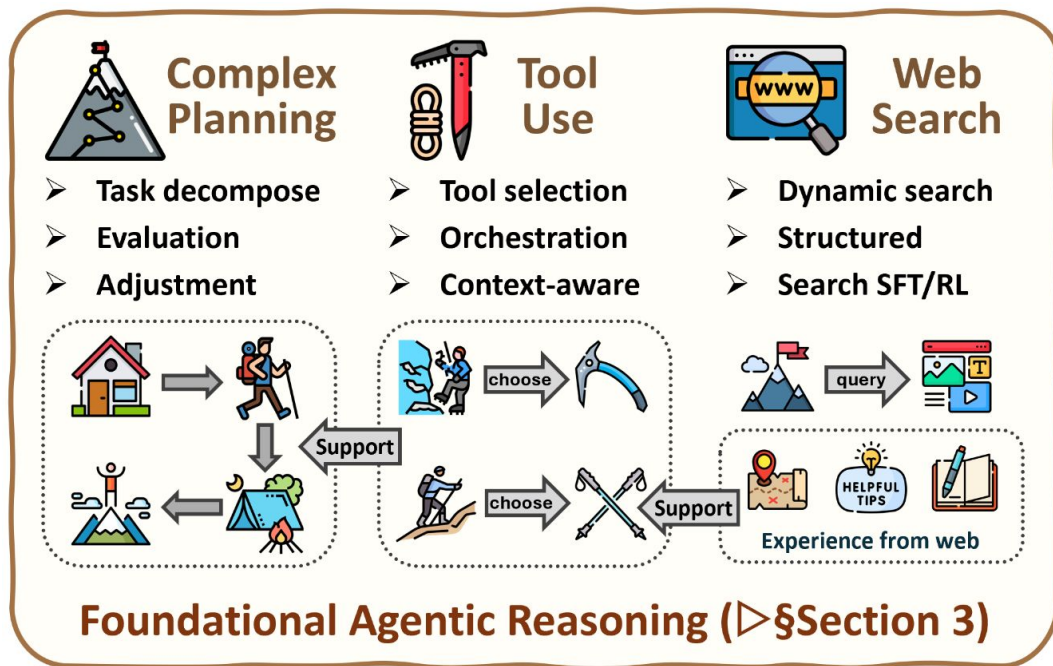
Decomposing complex problems into manageable steps.

Tool Use

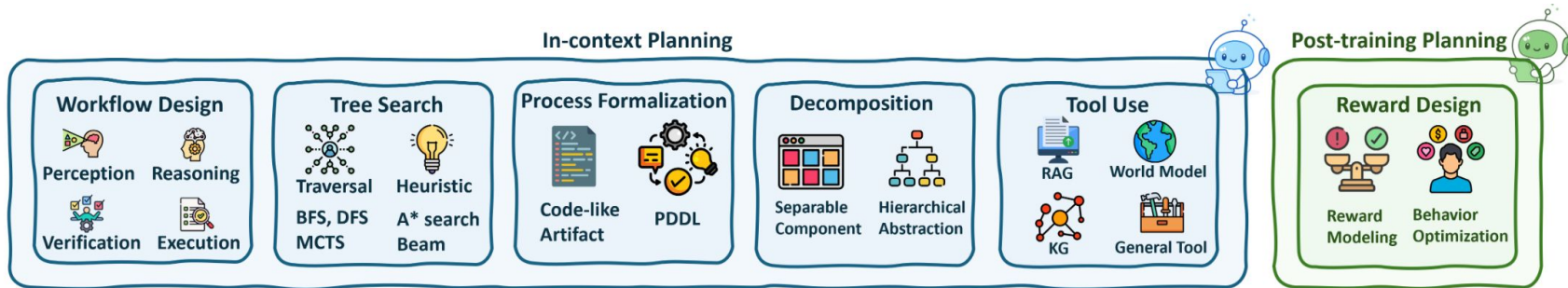
Augmenting intrinsic capabilities by invoking external modules (APIs, calculators, etc.).

Search

Extending support beyond the model's internal knowledge with dynamic context-aware searching.



Planning: Decompose Goals and Navigate Decision Spaces



In-Context Planning

- **Workflow Design**: Structured perception-reasoning-execution pipelines (e.g., ReAct).
- **Tree Search**: BFS/DFS/MCTS-style exploration of reasoning paths (e.g., Tree-of-Thoughts).
- **Process Formalization**: Code-like artifacts and PDDL-based planning.
- **Decomposition**: Hierarchical abstraction strategies.

Post-Training Planning

Reward Design: Internalizing planning strategies via RL-based optimization and behavior modeling.

"Planning is a central component of intelligent behavior, enabling agents to decompose problems, sequence decisions, and navigate complex environments with foresight."

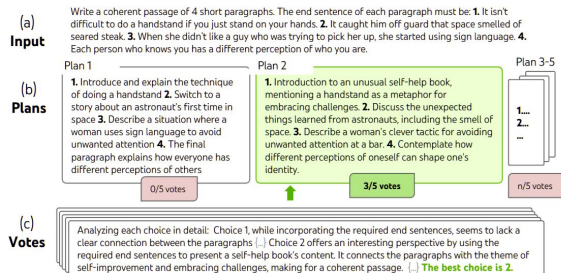
Planning Taxonomies

Method	Structure	Format	Tool
Modality I: Language Agents (e.g., Search Agents, Code Agents)			
ReWOO [71]	Decomposed	Natural Language	None
Reflexion [14]	Sequential	Natural Language	None
LLM+P [72]	Sequential	Formal Language	None
IPC [73]	Sequential	Formal Language	None
ToT [4]	Tree	Natural Language	None
GoT [74]	Graph	Natural Language	None
AoT [75]	Graph	Natural Language	None
HTP [76]	Hypertree	Natural Language	Retrieval
RefPlan [77]	Tree	Constrained Space	None
Gorilla [78]	Sequential	Programming Language	Retrieval, API
CodeNav [79]	Sequential	Programming Language	Code Indexer, Code Search
PoG [80]	Graph	Natural Language	Knowledge Graph
Tool-Planner [81]	Sequential	Natural Language	Tool Cluster
Modality II: Visual/Multimodal Agents (e.g., GUI Agents, Embodied Agents)			
VisualPredictor [82]	Tree	Formal Language	None
LLM-Planner [83]	Sequential	Formal Language	Object Detector, KNN
Agent-E [84]	Sequential	Formal Language	DOM Grounder, Screenshot
Agent S [85]	Hierarchical	Natural Language	API, Search, Memory
ExRAP [86]	Sequential	Natural Language	Memory
AESOP [87]	Reactive	Natural Language	Anomaly Detector
HRV [88]	Hierarchical	Formal Language	Symbolic Verifier
BehaviorGPT [89]	Sequential	Visual Features	World Model
Dino-WM [90]	Tree	Visual Features	World Model
FLIP [91]	Sequential	Visual Features	Language Model

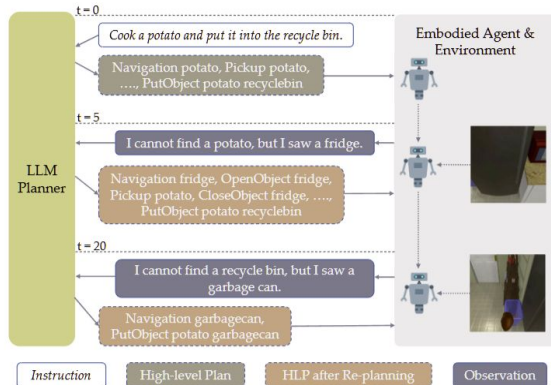
- **Structure** (planning organization)
 - Sequential:** step-by-step reasoning chain
 - Tree / Graph:** branching exploration over multiple reasoning paths
 - Hierarchical / Decomposed:** high-level plan decomposed into sub-tasks
- **Format** (representation of planning)
 - Natural Language:** planning expressed in text
 - Formal Language:** structured symbolic plans or programs
 - Programming Language:** executable code as plans
 - Visual / Feature Space:** planning over visual or multimodal representations

Planning Agents

- [1] Llm-planner: Few-shot Grounded Planning For Embodied Agents With Large Language Models
- [2] Tree Of Thoughts: Deliberate Problem Solving With Large Language Models
- [3] Gorilla: Large Language Model Connected With Massive Apis
- [4] A Goal Without A Plan Is Just A Wish: Efficient And Effective Global Planner Training For Long-horizon Agent Tasks



Language Plans [1]

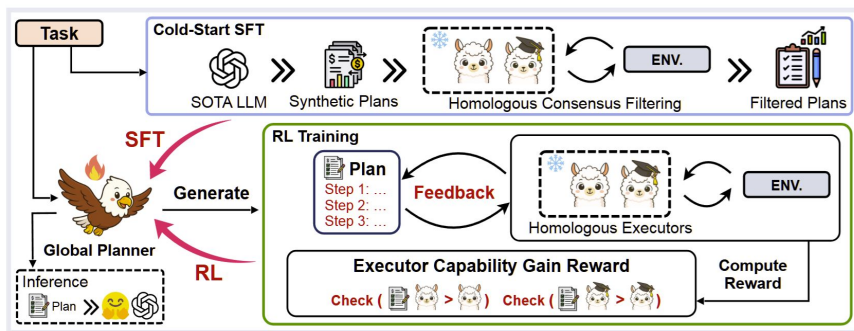


Symbolic Plans [2]

"Help me find an API to convert the spoken language in a recorded audio to text using Torch Hub."

Correct (Gorilla)	<pre><domain>: Speech-to-Text <api_provider>: TorchHub <code>: asr_model = torch.hub.load('snakers4/silero-models', 'silero_sst') result = asr_model.transcribe(audio_path)</pre>
API Argument Hallucination (GPT-4)	<pre><domain>: Speech-to-Text <api_provider>: TorchHub <code>: asr_model = torch.hub.load('snakers4/silero-models', 'asr', source='local') result = asr_model.transcribe(audio_path)</pre>
Incorrect API Call (Claude)	<pre><domain>: Audio-Translation <api_provider>: Pytorch <code>: import torchaudio translation = torchaudio.pipelines.WAV2VEC2_ASR_PIPELINE("audio.wav")</pre>

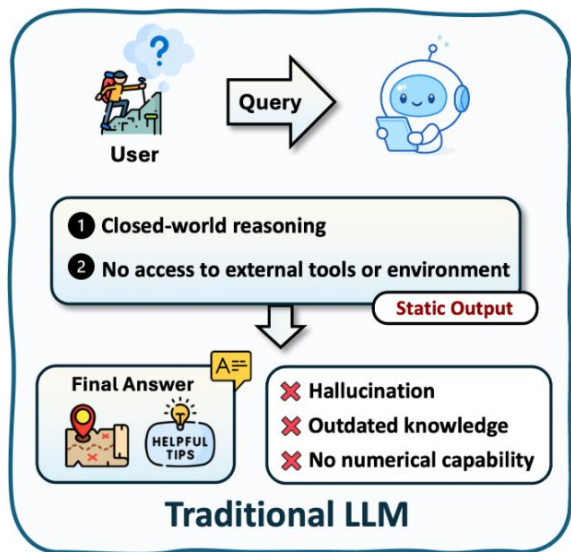
Coding Plans [3]/...



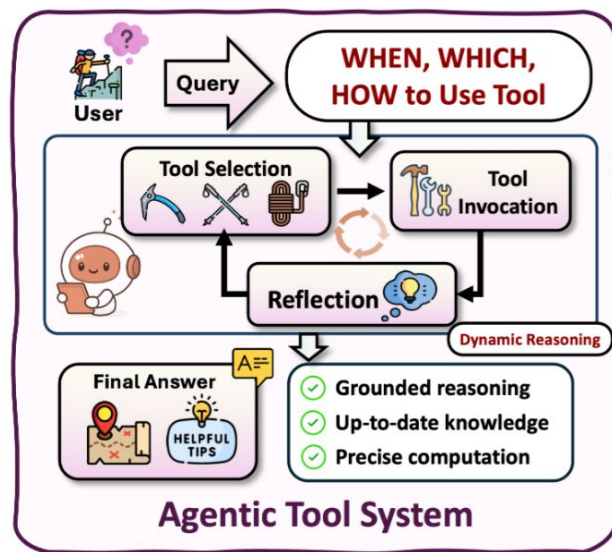
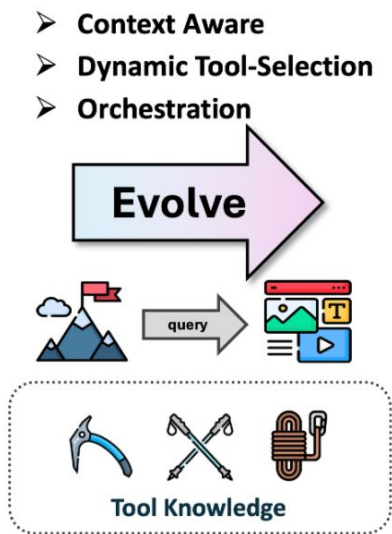
Post-training enhanced Plans [4]

Tool Use: from Chain-of-Thoughts to Chain-of-Actions

Traditional models operate in a closed world with fixed reasoning, agentic tool-use systems enable **dynamic selection, orchestration, and integration of external tools**, allowing agents to extend reasoning, improve precision, and dynamically adapt across domains



- Closed-world reasoning
- No access to external tools
- Prone to hallucination & outdated knowledge



- Dynamic tool selection & invocation
- Reflection and self-correction
- Grounded reasoning & precise computation

Tool-using Taxonomies

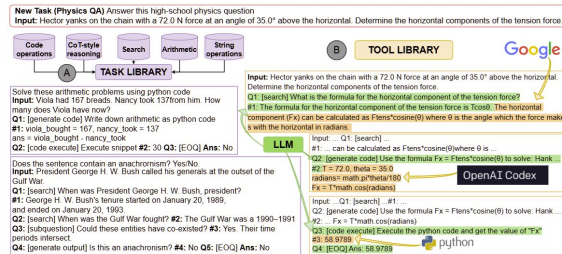
Method	Stage	Learning	Tool Strategy
Modality I: In-Context Integration			
ReAct [5]	Inference	Prompting	Interleaved reasoning–action
ART [199]	Inference	Few-shot	Retrieved multi-step demos
ChatCoT [200]	Inference	Prompting	CoT with tool calls
GEAR [201]	Inference	Delegation	Light model for tool selection
AVATAR [202]	Inference	Contrastive	In-context tool reasoning
Modality II: Post-Training Integration			
Toolformer [6]	Post-train	Self-sup. + SFT	Self-generated API calls
ToolLLM [203]	Post-train	SFT	Large-scale API demos
ToolAlpaca [204]	Post-train	SFT	Simulated dialogues
ReSearch [205]	Post-train	RL + Reflec.	Adaptive retrieval reasoning
ReTool [206]	Post-train	RL	Reinforced code execution
ToolRL [207]	Post-train	RL	Multi-tool policy learning
Modality III: Orchestration-based Integration			
HuggingGPT [7]	System	Planner–Exec.	Multi-tool coordination
TaskMatrix.AI [208]	System	Planner	Massive API ecosystem
ToolPlanner [81]	System	RL	Plan-before-act framework
OctoTools [209]	System	Rule-based	Hierarchical orchestration
ToolExpNet [210]	System	Embedding	Experience-based selection
ToolChain* [211]	System	Search	A* decision over tools

[1] ART: Automatic multi-step reasoning and tool-use for large language models

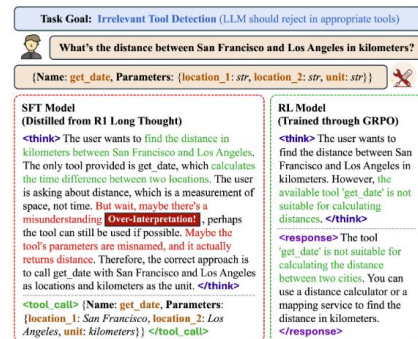
[2] ToolRL: Reward is All Tool Learning Needs

[3] OctoTools: An Agentic Framework with Extensible Tools for Complex Reasoning

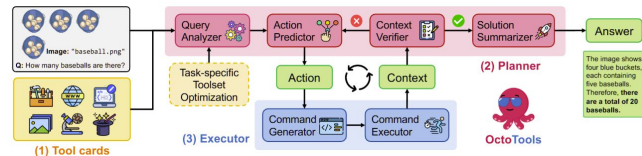
• In-Context Integration [1]



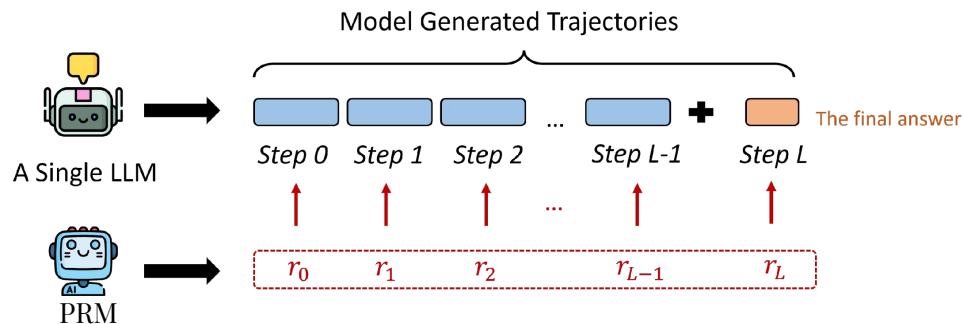
• Post-training Integration [2]



• Orchestration-based [3]



Process Reward Modeling (PRM) – “A Third Person View” for Agentic Scaling



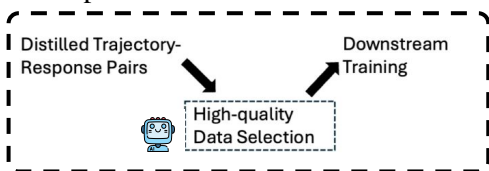
Utilization during **post-training**

- Select high quality data for SFT (example 1)
- Serve as the training reward signal in RL (example 2)

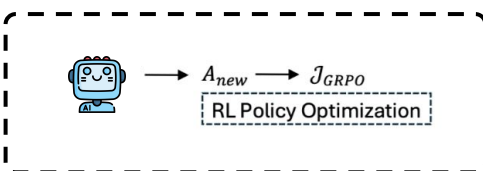
Utilization during **inference-time scaling**

- Best-of-N Sampling (example 3)
- Monte Carlo Tree Search
- Beam Search

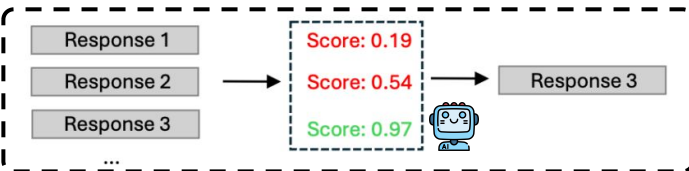
Example 1



Example 2



Example 3



[1] Zou, Jiaru, et al. "ReasonFlux-PRM: Trajectory-Aware PRMs for Long Chain-of-Thought Reasoning in LLMs." NeurIPS 2025

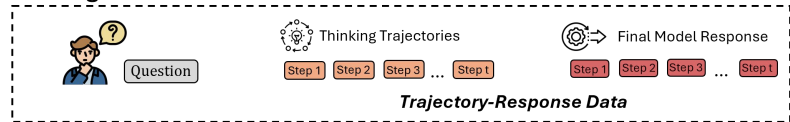
[2] Zhang, Zhenru, et al. "The lessons of developing process reward models in mathematical reasoning." arXiv 2025

PRMs for Agents and Tool Using

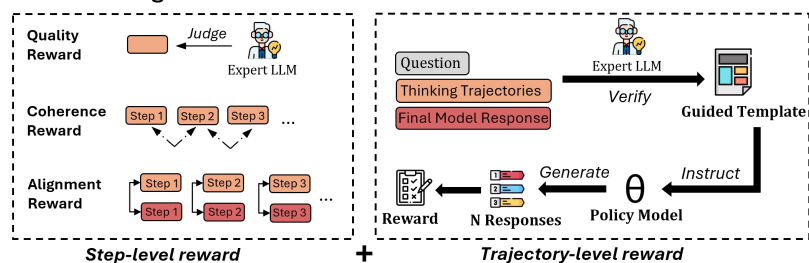
Discriminative PRM

- **Goal:** Learn a *scoring function* to rate reasoning steps or trajectories
- Step-wise **binary** labels (e.g. “[1,0]” or “[+,-]” for [correct, incorrect])
- Training as **classification tasks** with binary cross-entropy loss
- **Example:** ReasonFlux-PRM (Discriminative)

Training Data Curation



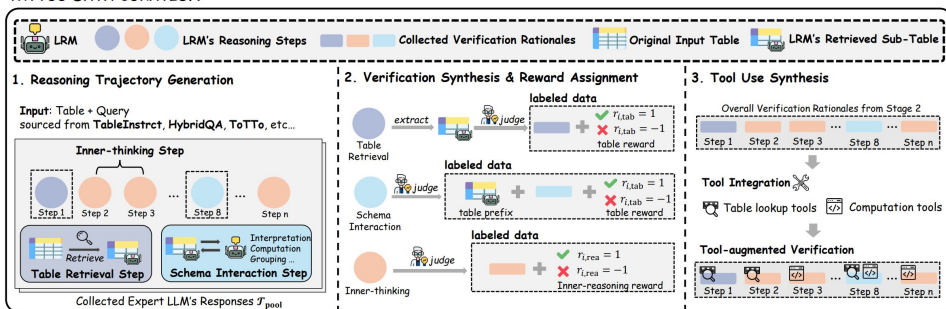
Reward Design



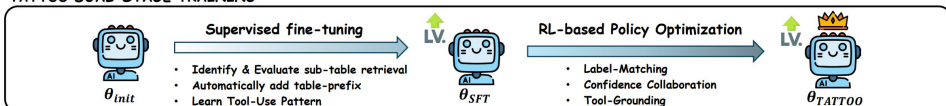
Generative PRM

- **Goal:** Generate *verification rationales* and *reward signals* for reasoning steps.
- Step-wise **CoT rationales** with **final judgement** as labels
- Training as **language modeling tasks** with supervised fine-tuning or RL fine-tuning
- **Example:** TaTTToo-PRM (Generative)

TATTOO DATA CURATION



TATTOO DUAL-STAGE TRAINING

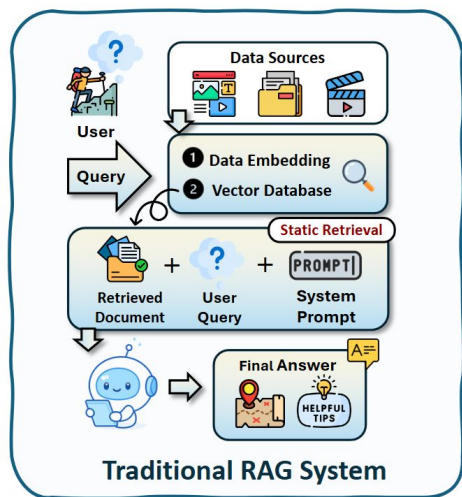


[1] Zou, Jiaru, et al. "ReasonFlux-PRM: Trajectory-Aware PRMs for Long Chain-of-Thought Reasoning in LLMs." NeurIPS 2025

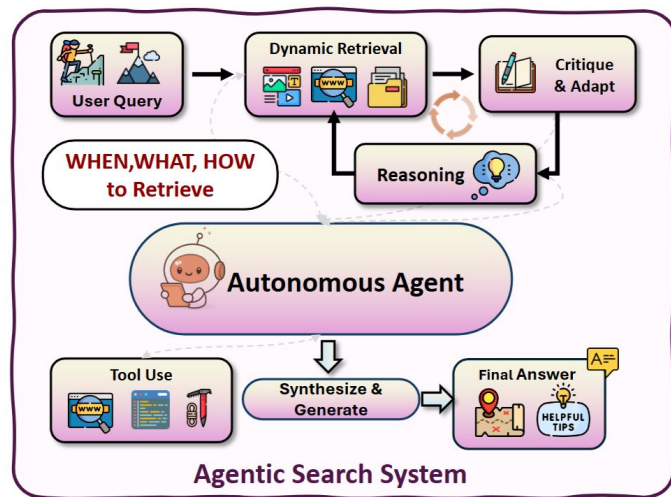
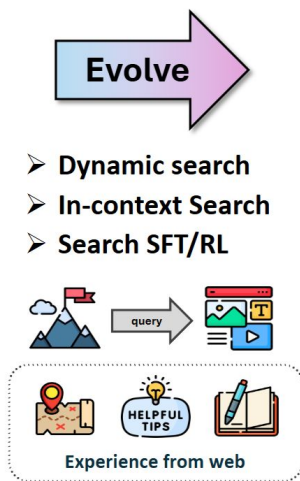
[2] Zou, Jiaru, et al. "TaTTToo: Tool-Grounded Thinking PRM for Test-time Scaling in Tabular Reasoning." ICLR 2026

Search: Extend beyond Internal Knowledge

Traditional RAG relies on static retrieval over a vector database, while agentic search introduces **autonomous decision-making for when, what, and how to retrieve**, enabling dynamic search, in-context retrieval, critique-and-adapt loops, and tool use.



- Static, one-time retrieval
- Fixed vector database search
- No adaptation during generation

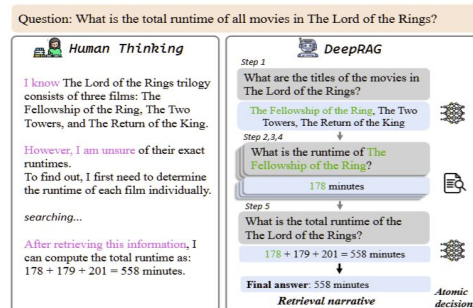


- Dynamic, multi-step retrieval
- Critique-and-adapt loop
- Autonomous decision-making

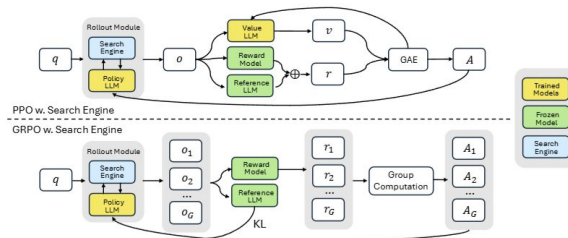
Search Taxonomies

Method	Structure	Format	Tool
Modality I: In-Context Agentic Search			
ReAct [5]	Interleaved	NL + Actions	Search API
Self-Ask [254]	Decomposed	NL Queries	Search API
IRCoT [213]	Sequential	NL + CoT	Search API
Self-RAG [255]	Reflective	NL Self-check	Conditional Search
DeepRAG [256]	Iterative	NL Feedback	Search API
Modality II: Post-Training Agentic Search			
Toolformer [6]	Sequential	Tool Tokens	APIs, Search
INTERS [257]	Sequential	Instructions	Search API
WebGPT [258]	Sequential	NL + Browser	Web Search
RAG-RL [259]	Decision	NL Policy	Evidence API
Search-R1 [27]	Iterative	NL + Tokens	Live Web
Deep-Researcher [260]	Multi-step	NL Trajectories	Browser Tools
ReSearch [205]	Step-wise	NL Steps	Search + Verifier
ReARTer [261]	Reflective	NL Policy	Tool Cluster
Modality III: Structure-Enhanced Agentic Search			
Agent-G [262]	Modular	NL + Graph Ops	KG Query
MC-Search [263]	Multi-step	NL	Multimodal Search
GeAR [264]	Graph	Graph Ops	KG Expansion
ARG [265]	Reflective	NL + Symbols	KG Traversal

- **In-Context Agentic Search [1]**



- **Post-training Agentic Search [2]**



- **Structure-enhanced Agentic Search [3]**



[1] DeepRAG: Thinking to Retrieve Step by Step for Large Language Models

[2] Search-R1: Training LLMs to Reason and Leverage SearchEngines with Reinforcement Learning

[3] MC-Search: Evaluating and Enhancing Multimodal Agentic Search with Structured Long Reasoning Chains

Agent Environment Requires Structured Reasoning

Queries for MLLMs are often ambiguous and complex

- require multi-step, cross-modal, and knowledge-intensive reasoning

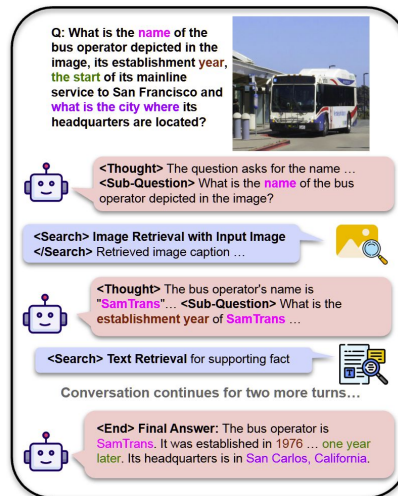
Demands richer agentic behaviors and adaptive reasoning

- iterative task decomposition,
- adaptive cross-modal retrieval
- multimodal evidence integration
- ...

Current Benchmarks

Table 1: **Left:** Comparison of existing multimodal retrieval-augmented QA datasets. **Right:** Distribution of the five reasoning topologies in MC-SEARCH, with outer rings illustrating hop diversity (2–5 hops).

Benchmark	Knowledge Modality	Typical Hops	Long Chain (≥ 4 Hops)	Stepwise Annotation	Reasoning Topology
OK-VQA (Marino et al., 2019)	Text	1 Hop	✗	✗	✗
ViQuAE (Lerner et al., 2022)	Text	1 Hop	✗	✗	✗
WebQA (Chang et al., 2022)	Text/Caption	≤2 Hops	✗	✗	✗
InfoSeek (Chen et al., 2023)	Text	1 Hop	✗	✗	✗
MMSearch (Jiang et al., 2024)	Text/Image	1 Hop	✗	✗	✗
Dyn-VQA (Li et al., 2025b)	Text/Image	≤2 Hops	✗	✗	✗
MRAG (Hu et al., 2025)	Image	1 Hop	✗	✗	✗
M ² RAG (Liu et al., 2025c)	Text/Image	1 Hop	✗	✗	✗
MC-SEARCH	Text/Image	≥4 Hops	✓	✓	✓



Most datasets:

- 1–2 hops
- fixed retrieve-then-generate

no step-wise annotations
no reasoning topology modeling

What is MC-Search?

The First Benchmark for Agentic Multimodal RAG with Long Structured Chains

- 3,333 high-quality samples
- Avg 3.7 hops
- 5 reasoning topologies
- Stepwise gold chains
- Hop-wise attribution and verification filtering

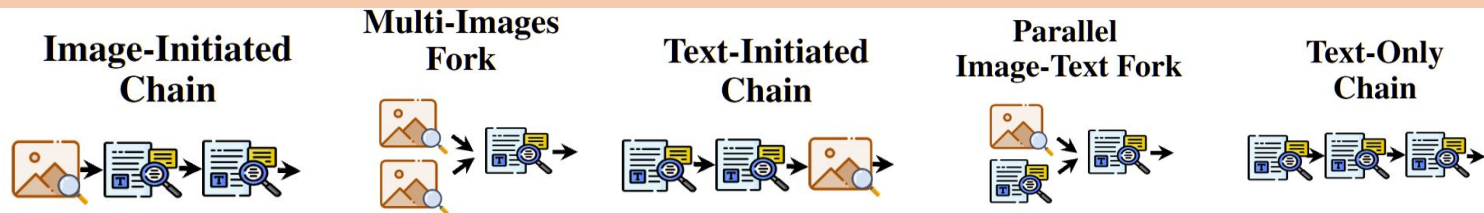
Improved Context Utility:

$$Util(t) = F1(C) - F1(C \setminus r_t)$$

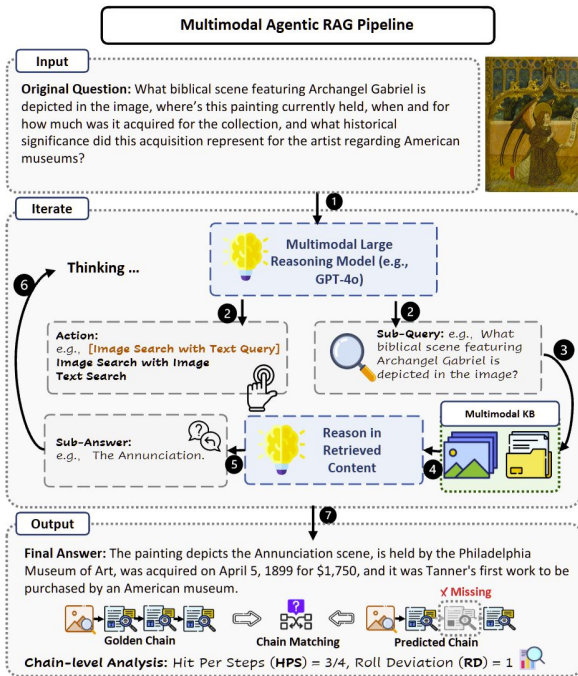
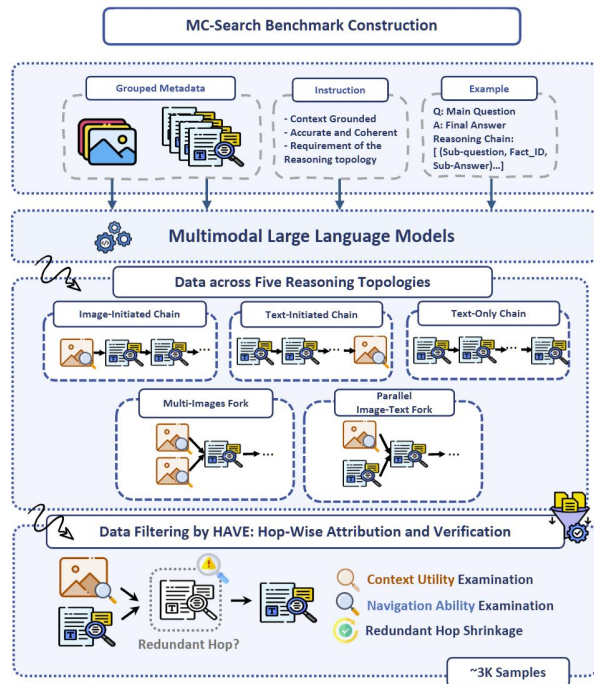
Highlight Navigation Role:

$$Nav(t) = 1 \text{ if entity propagates downstream}$$

Statistic	Number
Parallel Image-Text Fork	680
Image-Initiated Chain	1,306
Text Chain	945
Text-Initiated Chain	169
Multi-Images Fork	233
Total Samples	3,333
KB Images	389,750
KB Documents	784,473
Avg. Hops	3.79



Why Final Answer Accuracy is Not Enough



Step-wise Annotations:

- Retrieval errors may cancel out
- Over-retrieval hidden
- Planning mistakes are invisible
- We need process-level metrics.

Chain metrics:

- HPS (Hit per Step)
- RD (Rollout Deviation)

$$\text{HPS} = \text{Hit Docs} / \text{Retrieved Docs}$$
$$\text{RD} = ||\text{Golden Chain}| - |\text{Actual Chain}||$$

Agentic Evaluation Pipeline

- Multimodal reasoning chains for structured evaluation
- **SEARCH-ALIGN**, trained on our dataset, consistently improves agent performance

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Next

Pillar 2: Self-Evolving Agentic Reasoning

This layer addresses an agent's capacity to improve its own reasoning process through **past experience**. Two fundamental mechanisms drive this evolution: **feedback** and **memory**. Together, they transform reasoning from a static process into a dynamic, adaptive loop capable of **continual improvement of agentic capabilities** (planning/tool-use/search, etc.).

Feedback

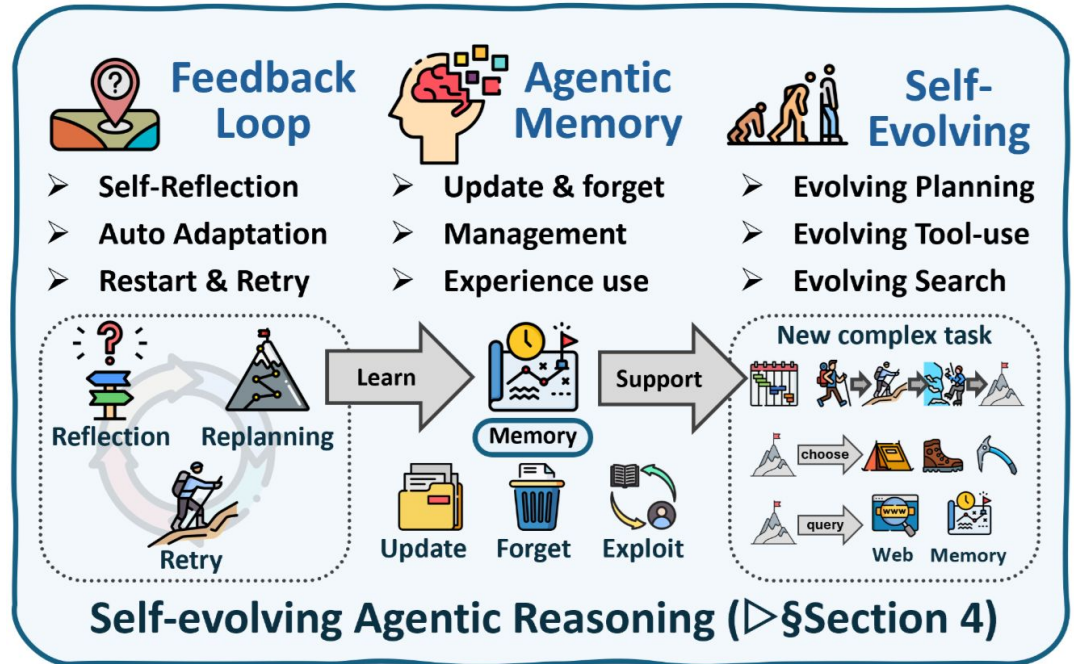
Iteratively refine their reasoning and actions rather than relying on one-shot responses.

Memory

Storing and organizing past interactions, enabling knowledge accumulation and reuse across tasks.

Self-evolving

Continual improvement of agentic capabilities (planning/tool-use/search, etc.).

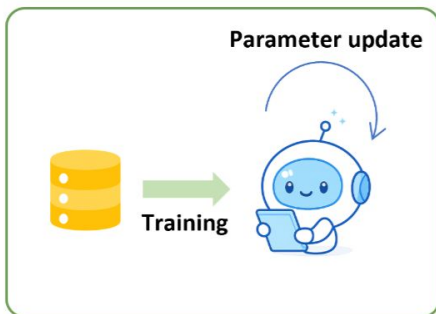


Self-Evolving: Agentic Feedback Mechanisms

Reflective Feedback



Parametric Adaptation



Validator-Driven Feedback

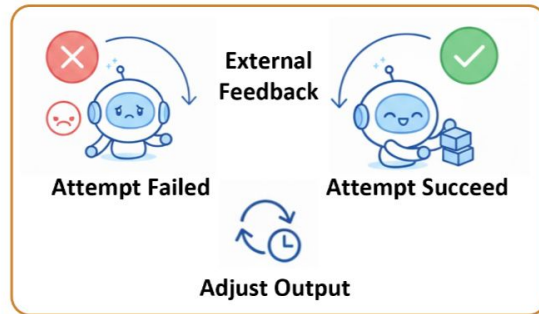


Figure 5: Agentic Feedback Mechanisms

Reflective Feedback

INFERENCE-TIME SELF-CRITIQUE

The agent generates a reasoning trajectory, critiques its own output, and revises it immediately.

Example: Reflexion, Self-Critique

Parametric Adaptation

OFFLINE LEARNING

Feedback signals are consolidated into parameter updates via fine-tuning or RL, internalizing lessons.

Example: RLHF, Fine-tuning on failures

Validator-Driven

OUTCOME-BASED RETRY

External validators (compilers, tests) provide pass/fail signals. The agent adjusts and retries on failure.

Example: Code generation with unit tests

Self-Evolving: Rewards are the Key

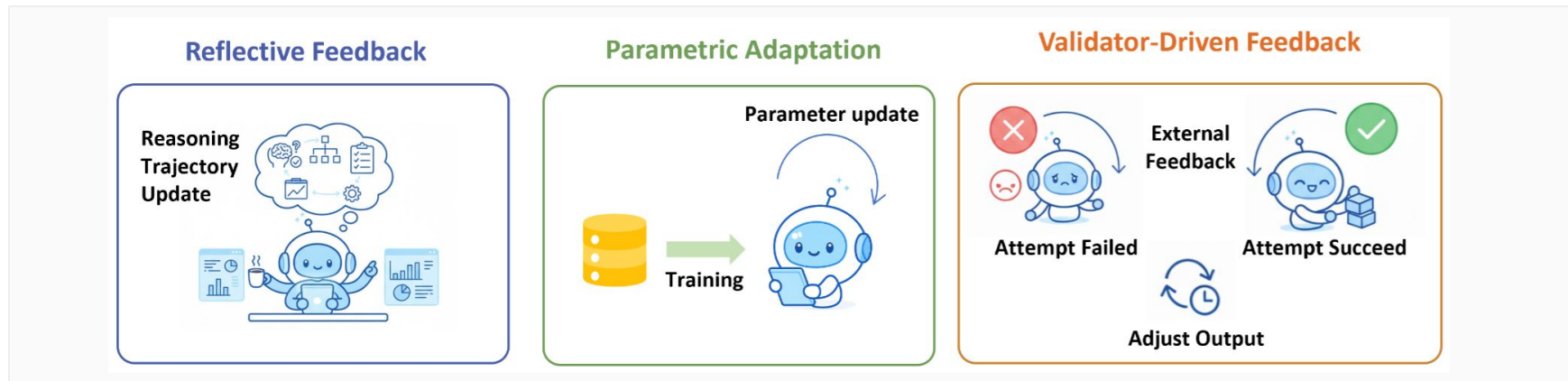


Figure 5: Agentic Feedback Mechanisms

Environment Feedback

ENVIRONMENT INTERACTION

Rewards obtained from **environment interaction**.

- task success/failure
- simulator reward
- executor feedback in embodied agents

Programmatic Feedback

AUTOMATED RULES

Feedback computed by **automated evaluation rules**.

- unit tests for coding tasks
- execution correctness
- rule-based verification

Human Feedback

HUMAN ANNOTATIONS

Feedback provided by **human annotations**.

- QA correctness labels
- preference comparisons
- human evaluation

Model-based Feedback

MODEL-EVALUATION

Feedback generated by **another model or the same model**.

- LLM judge
- self-reflection reward
- critique models

Self-Evolving: The Role of Agentic Memory

Agentic memory transforms agents from stateless processors into experience-aware reasoners, organized along three parallel dimensions:

In-Context Use

Leveraging the context window for short-term recall. Includes **Conversation Memory** (raw history & summaries) and **Experience Memory** (workflows & reasoning paths).

Structured Representation

Persistent, organized long-term storage. **Graph Memory** connects entities and facts; **Multimodal Memory** integrates text, images, and sensory data.

Post-Training Control

Learned memory management via **Reward-Guided Mechanisms**. Agents actively decide what to store, update, or forget to optimize future reasoning.

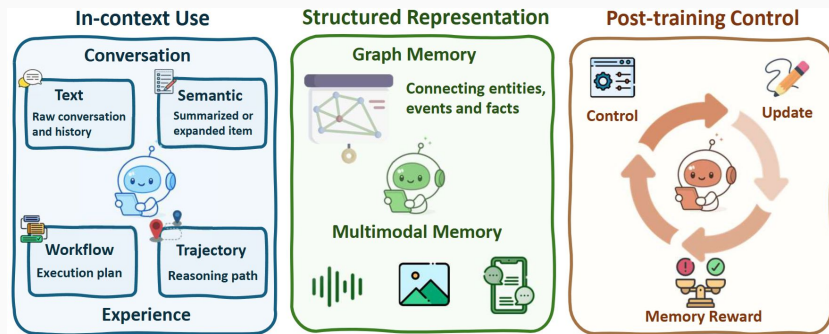


Figure 6: Overview of **Agentic Memory** in LLM agents, showing three parallel dimensions: in-context use (text and experience), structured representation (graph and multimodal memory), and post-training control (reward-guided memory management).

User Task: Personal Memory

Real-World Conversation with MLLM Agents:

- Multi-session
- Information accumulates, evolves
- Vision + language are jointly used

Existing Benchmarks:

- Text-only multi-session memory
- Multimodal single-session context understanding

Mem-Gallery: New Conversation Dataset + Evaluation Framework

- Multi-session conversations with non-trivial multimodal dependency
- Unified evaluation framework for three key memory capabilities

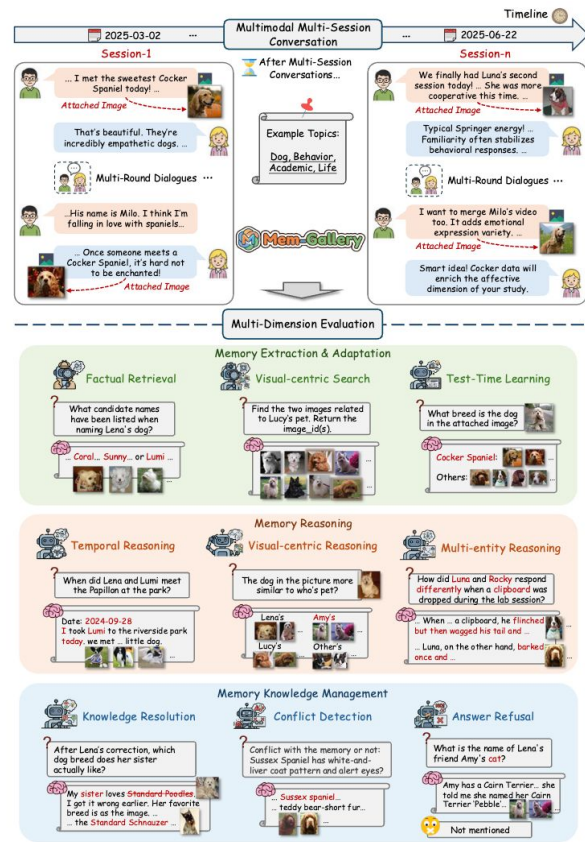


Figure 1: The conceptual illustration of Mem-Gallery.

Characteristics

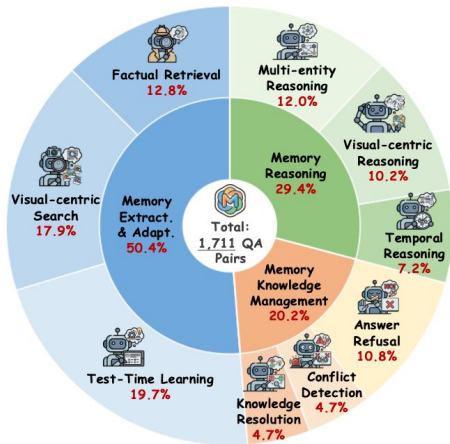


Table 1: Comparison between Mem-Gallery with representative conversational benchmarks. ✓: Satisfies; ✗: Does not satisfy; ✎: Text modality only.

Benchmark	Conversational Characteristics				Extract.&Adapt.			Reasoning			Management		
	A. Round	A. Img.	Multi-Sess.	MM Info.	FR	VS	TTL	TR	VR	MR	KR	CD	AR
DuLeMon (Xu et al., 2022)	8.16	—	✓	✗	✎	✗	✗	✗	✗	✗	✗	✗	✗
DialogBench (Ou et al., 2024)	7.48	—	✓	✗	✎	✗	✗	✗	✗	✗	✗	✗	✗
MemoryBank (Zhong et al., 2024)	3.77	—	✓	✗	✎	✗	✗	✗	✗	✗	✗	✗	✗
MMDU (Liu et al., 2024)	14.95	3.83	✗	✓	✓	✗	✗	✗	✓	✗	✗	✗	✗
LoCoMo (Maharana et al., 2024)	10.81	3.35	✓	✓	✓	✗	✗	✗	✓	✗	✗	✗	✗
LOCCO (Jia et al., 2025)	4.77	—	✓	✗	✎	✗	✗	✗	✗	✗	✗	✗	✗
LongMemEval (Wu et al., 2025)	5.19	—	✓	✗	✎	✗	✗	✗	✗	✗	✗	✗	✗
MemoryAgentBench (Hu et al., 2025a)	9.55	—	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
MMRC (Xue et al., 2025)	12.90	2.90	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓
Mem-Gallery (Ours)	16.51	4.18	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

* **A. Round**: average user-assistant dialogue round numbers per session. **A. Img.**: average image numbers per session. **Multi-Sess.** & **MM Info.**: whether the dataset has basic characteristics of multi-session and multimodal information. This property serves as a fundamental prerequisite for multimodal long-term conversational memory. For the three evaluation dimensions: (i) **Memory extraction & adaptation** includes factual retrieval (FR), visual-centric search (VS), and test-time learning (TTL) subtasks. (ii) **Memory reasoning** includes temporal reasoning (TR), visual-centric reasoning (VR), and multi-entity reasoning (MR) subtasks. (iii) **Memory knowledge management** includes knowledge resolution (KR), conflict detection (CD), and answer refusal (AR) subtasks.

Memory Extraction & Adaptation:

- Factual Retrieval
- Visual-centric Search
- Test-time Learning

Memory Reasoning:

- Temporal Reasoning
- Visual-centric Reasoning
- Multi-entity Reasoning

Memory Knowledge Management:

- Knowledge Resolution
- Conflict Detection
- Answer Refusal

Advantages of Mem-Gallery



Comprehensive Comparisons with unified pipeline

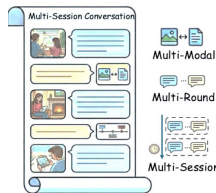
Compared with the representative work LoCoMo:

Visual Information Effectiveness



Figure 2: Effectiveness analysis of visual information with representative memory models. Compared with LoCoMo ([Maharana et al., 2024](#)), visual information plays a much more critical role in Mem-Gallery.

Average 198.1 Rounds Per Conversation
Average 12.0 Sessions Per Conversation



High-quality Long-term data

Image Number Distribution (per Session)

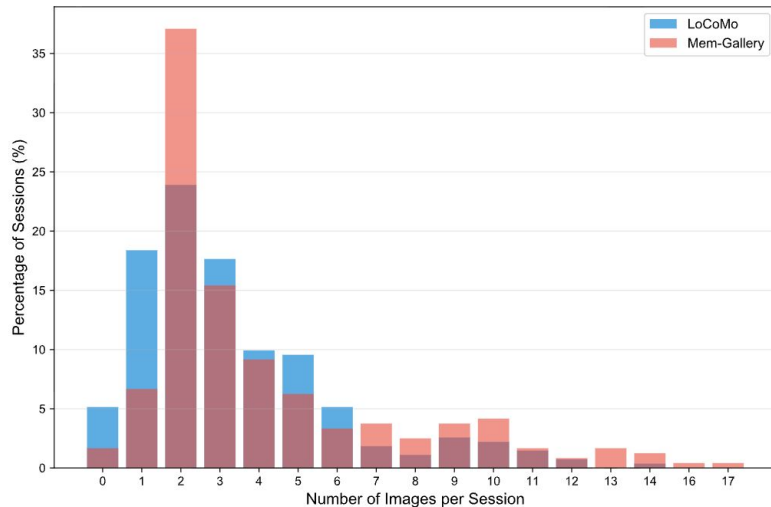
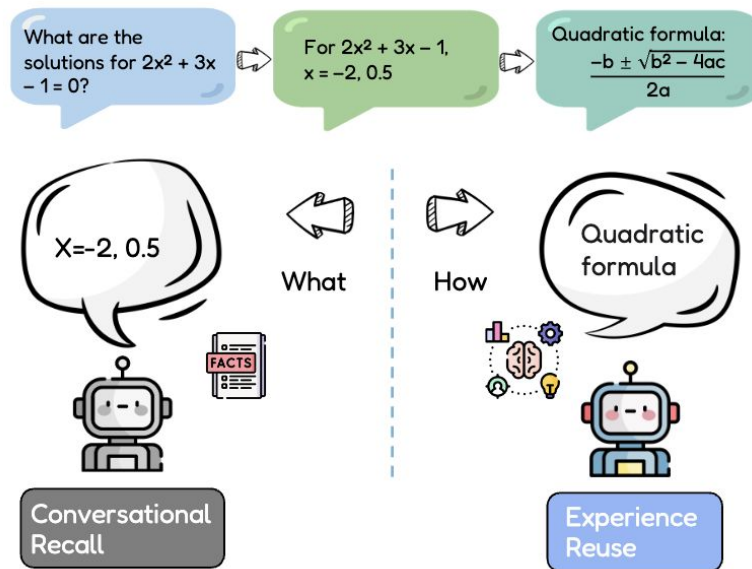
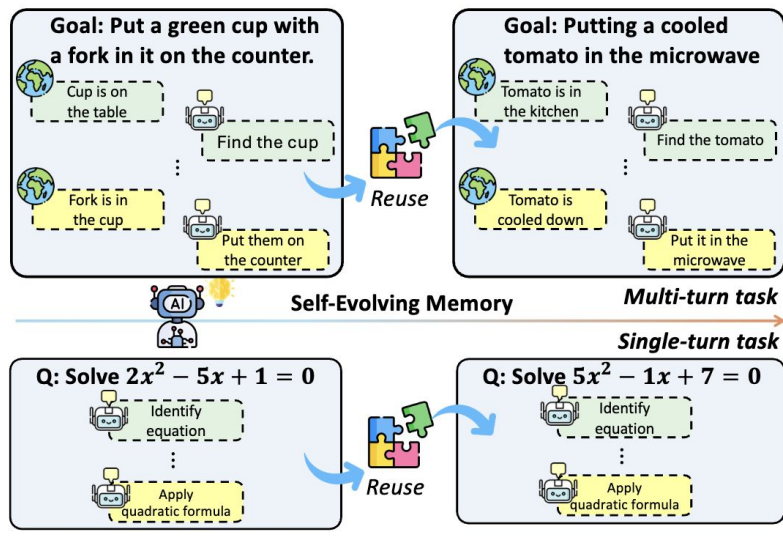


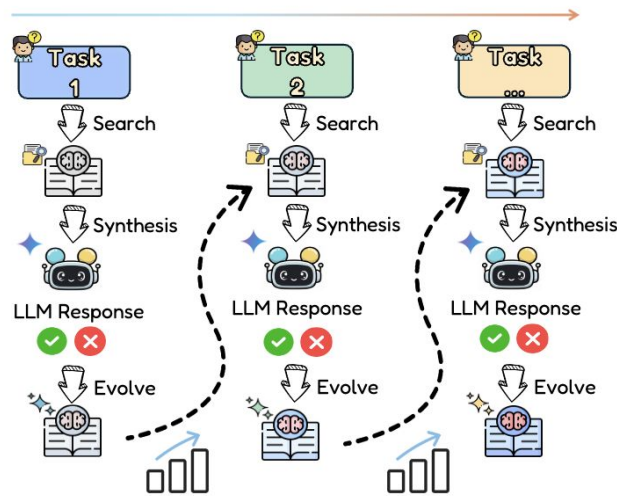
Figure 8: Image distribution of Mem-Gallery per conversation session compared with LoCoMo.

Agent Task: Experience Memory

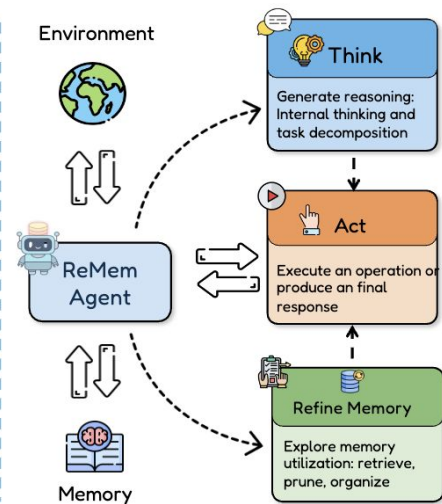


- Most agent memory today focuses on **conversational recall**.
- This means recalling *what was said*, not *what was learned*.
- Real agents need to reuse experience and improve over time.

Problem Setting: Streaming and Evolving Tasks



(a) Test-time Evolution Setting

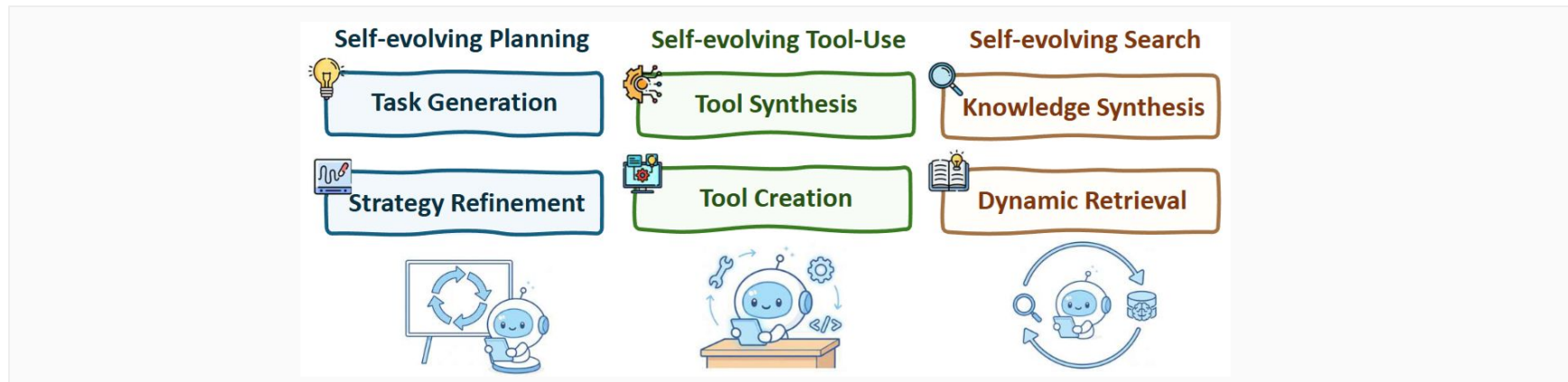


(b) Illustrative Think-Act-Refine Loop

- We consider tasks arriving as a continuous stream.
- Tasks can be single-turn reasoning or multi-turn interaction.
- The key challenge is learning across tasks at test time.

- Proposed ReMem, which has three core operations. Think, Act, and Refine Memory.
- Unlike prior agents, memory itself becomes a reasoning object.
- Reorganize and prune memory during inference.

Self-Evolving: Advancing/Creating Agentic Skills



Evolving Planning

Synthesizing Planning Tasks

Agents can autonomously generate tasks, learn from their own feedback, and adapt strategies through iterative interaction with the environment.

Example: SELF, SCoRe, PAG, TextGrad

Evolving Tool-use

Creating and Synthesizing Tools

The culmination of in-context reasoning is the emergent capability of agents to autonomously create new tools.

Example: LATM, CRAFT

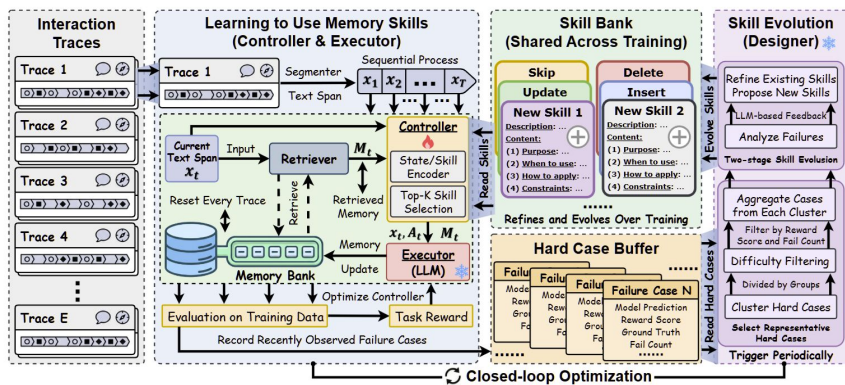
Evolving Search

Evolving Memory Bases

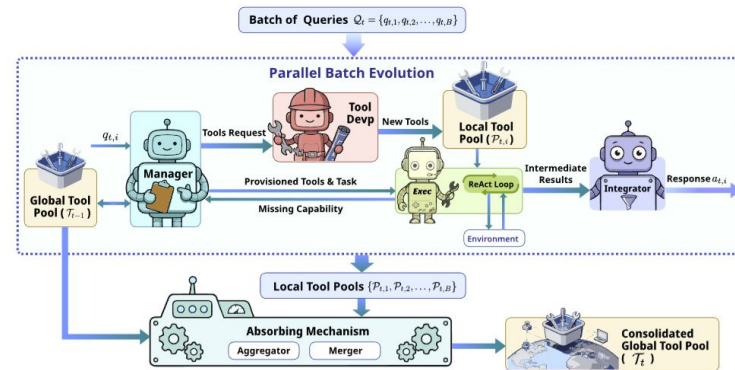
Agents continuously update their memory base during task execution, while dynamically adjusting how search is performed over this evolving knowledge.

Example: MemOS, MemGPT

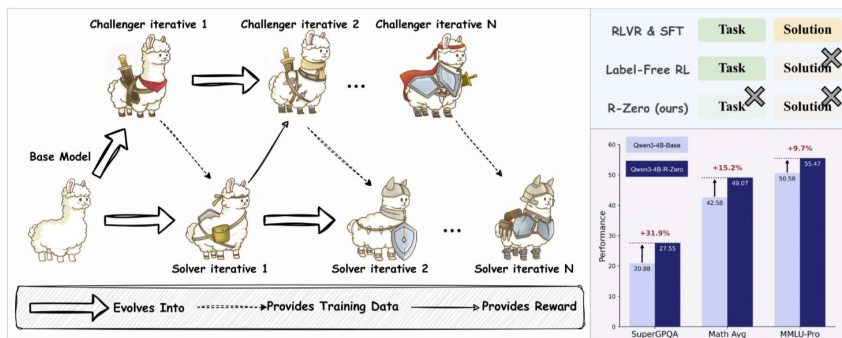
Self-Evolving: Advancing/Creating Agentic Skills



Evolving Skills [1]



Evolving Tools [2]



Data and Model Co-evolving [3]

To be evolved...

[1] MemSkill: Learning and Evolving Memory Skills for Self-Evolving Agents

[2] Yunjue Agent Tech Report: A Fully Reproducible, Zero-Start In-Situ Self-Evolving Agent System for Open-Ended Tasks

[3] R-Zero: Self-Evolving Reasoning LLM from Zero Data

Roadmap

- Background and Overview
- Pillar 1: Foundational Agentic Reasoning
- Pillar 2: Self-Evolving Agentic Reasoning
- **Pillar 3: Collective Multi-Agent Reasoning**
- Applications and Benchmarks
- Open Challenges
- Discussion



Pillar 3: Collective Multi-Agent Reasoning

Rather than identical problem solvers, **agents assume complementary roles**, such as Manager for task decomposition, Worker for execution, and Verifier for evaluation, enabling specialization and division of cognitive labor. This role differentiation marks the first step toward collective intelligence, where **reasoning is distributed and coordinated across multiple agents**.

Role Assigning

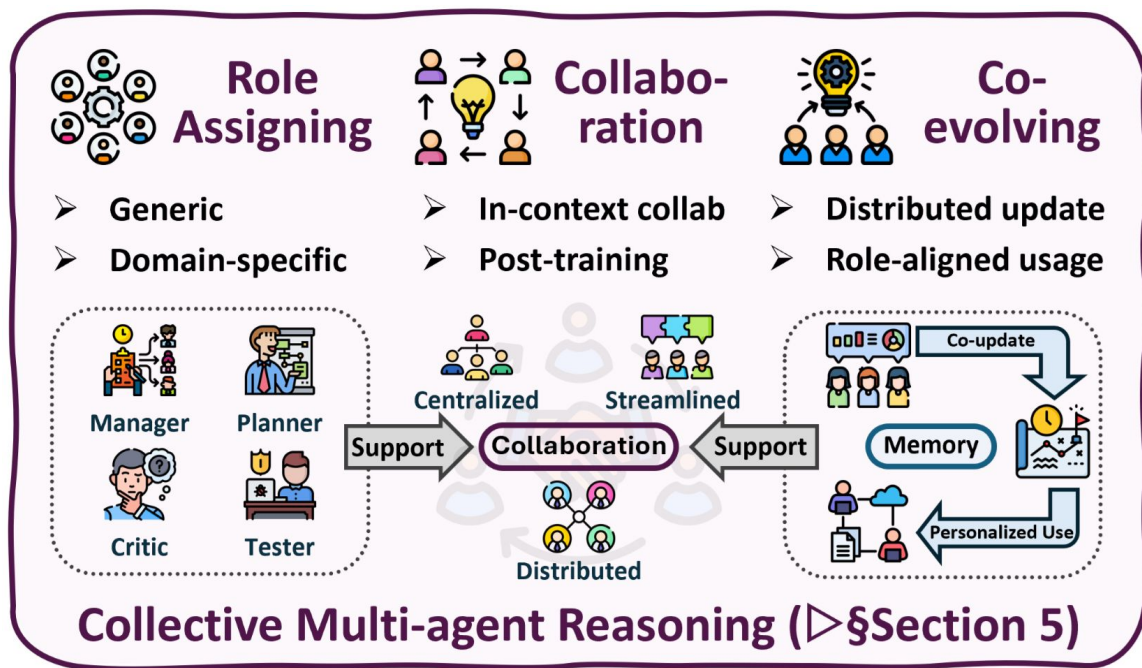
How to design static or adaptive roles that align with task structure and expertise distribution.

Collaboration

How agents exchange intermediate reasoning, negotiate consensus, and divide labor efficiently.

Co-evolving

How shared or distributed state supports long-term coordination and continual adaptation.



Multi-Agent Systems: Generic Role Taxonomy

Generic Roles

Leader	Sets objectives, decomposes tasks
Worker	Executes assigned subtasks
Critic	Reviews outputs, identifies errors
Memory Keeper	Maintains shared context
Facilitator	Manages information flow

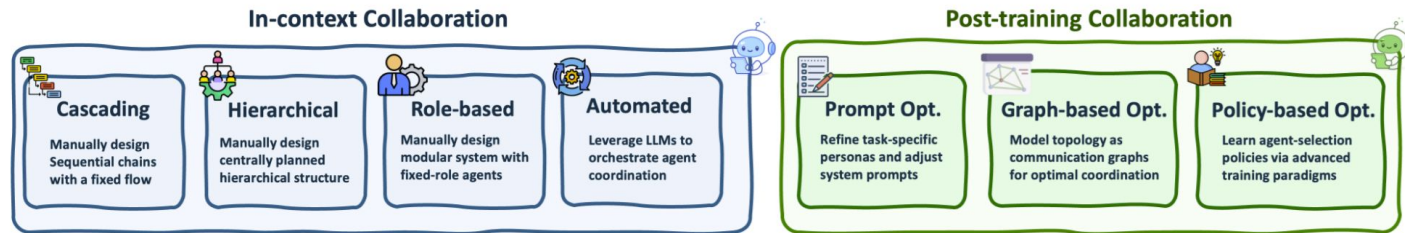
Domain-Specific Roles

Software Engineering	Healthcare & Medicine
Finance & Analysis	Legal Research

Multi-agent systems often mirror organizational structures by assigning generic roles. These roles distribute cognitive labor, creating a system of checks and balances.



Collaboration and Division of Labor



In-Context Collaboration

Coordination strategies are specified at inference time without additional training.

- **Manually Crafted Pipelines:** Relies on predefined hierarchies or fixed workflows (e.g., MetaGPT).
- **LLM-Driven Pipelines:** Leverages an LLM as an orchestrator to dynamically decompose goals and route tasks.

Post-Training Collaboration

Optimizes agent roles, interaction structures, or routing policies through learning or search.

- **Prompt Optimization:** Refines agent roles and workflows encoded in prompts to yield better coordination.
- **Topology Optimization:** Treats the MAS as a graph and uses graph-based or policy-based methods to learn an optimal communication structure.

The Four Dimensions of Multi-Agent Memory

Four Dimensions of Memory Design

Architecture

Hierarchical (global coherence)
vs. Flat (simplicity).

Content

Semantic (concepts) vs.
Procedural (action sequences).

Topology

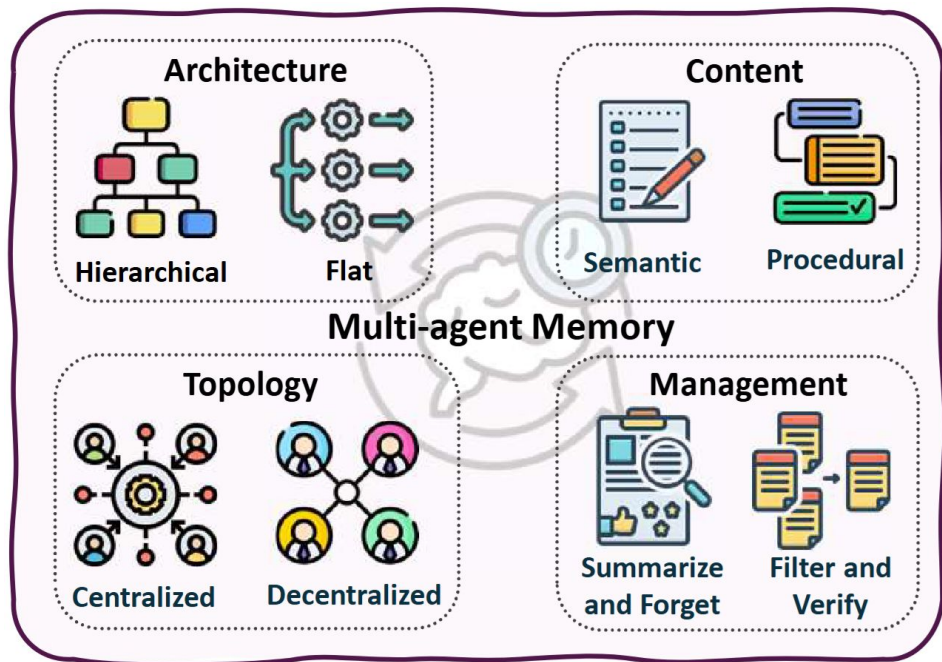
Centralized (consistent) vs.
Decentralized (scalable).

Management

Summarize & Forget vs. Filter &
Verify mechanisms.

Collaboration Strategies

- **In-Context Collaboration:**
Agents share reasoning through structured message passing.
- **Post-Training Collaboration:**
Agents are jointly trained with shared reward signals (MARL).



Training Multi-Agent Systems to Evolve

Recent advancements are shifting multi-agent systems from fixed, hand-designed coordination toward training paradigms that enable agents to evolve collectively over time. This transforms the system into a **self-improving organization** that **adapts its structure, communication, and policies** in response to the environment.

Co-evolution via Interaction

Utilizes training objectives that couple interaction, feedback, and role specialization, allowing for collective skill growth without external supervision.

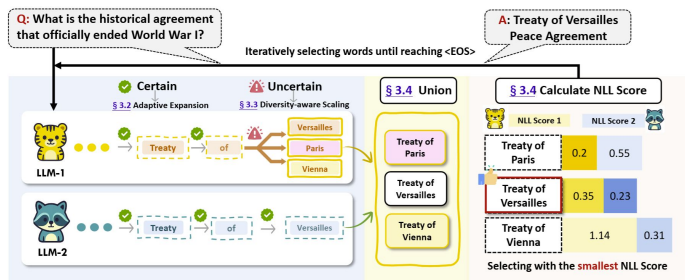
Multi-Agent Reinforcement Fine-Tuning

Applies principled RL frameworks tailored to LLM-based agent organizations to stabilize collective post-training adaptation.

Preference- and Alignment-Driven Evolution

Studies evolution under preference- and alignment-driven objectives, learning collective policies from human-derived or AI-judged feedback.

Collective Multi-Agent Reasoning



Multi-agent Context Inference [1,2,3]

- Combine multiple LLM outputs to reduce uncertainty and improve answer reliability

- AdaFuse: Adaptive Ensemble Decoding with Test-Time Scaling for LLMs
- Harnessing Consistency for Robust Test-Time LLM Ensemble
- Inference Scaling of LLM Ensembling: Bridging Token Spaces with Token Translation

Multi-agent Collective Memory [4]

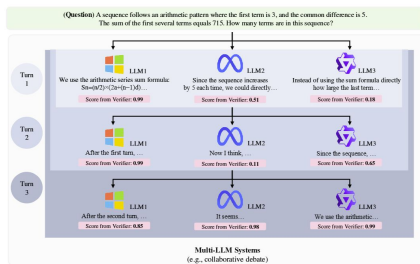
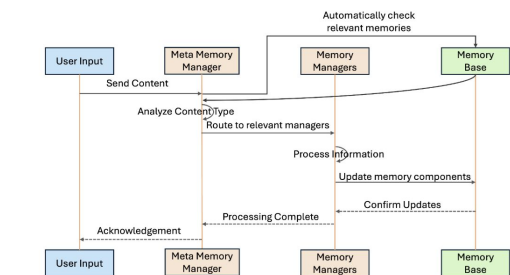
- Structured memory management enables persistent context and long-term reasoning

- MIRIX: Multi-Agent Memory System for LLM-Based Agents

Multi-agent Collective Training [5,6]

- Multiple agents coordinate through iterative interaction and verification
- Improve reasoning through collaborative optimization

- MAPoRL2: Multi-Agent Post-Co-Training for Collaborative Large Language Models with Reinforcement Learning
- Transformer Copilot: Learning from The Mistake Log in LLM Fine-tuning



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Applications: Agentic reasoning is transforming scientific discovery, healthcare, robotics, and software engineering

Math & Code

Symbolic manipulation, iterative debugging, and collaborative proof construction.

AlphaProof, Manus AI, Cursor, GitHub Copilot, Codex and Claude Code

Scientific Discovery

Hypothesis generation, experiment automation, and literature synthesis.

AI Scientist, DeepResearch

Embodied Agents

Perception-grounded planning, physical action execution, and environment adaptation.

LLM-Planner, Gemini Robotics, Octopus

Healthcare & Medicine

Clinical reasoning, diagnosis support, and personalized treatment planning.

MedAgent, ClinicalAgent, MedOrch, AgentMD

Autonomous Research

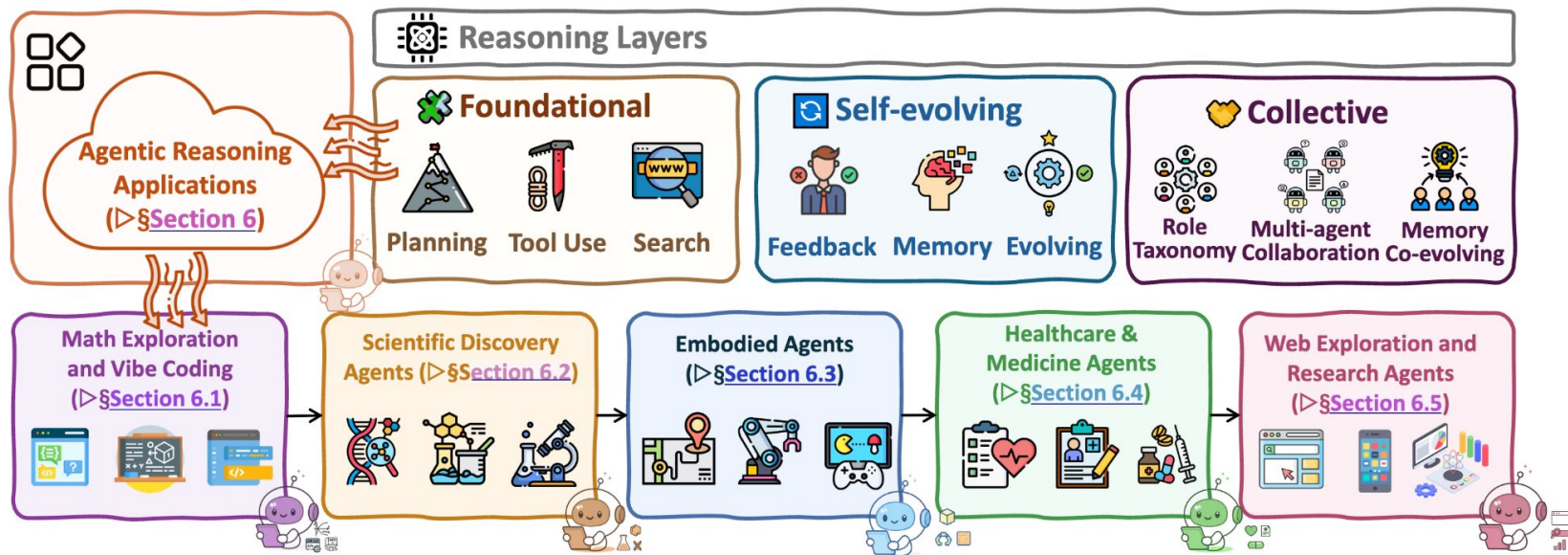
Long-horizon web navigation, information synthesis, and report generation.

WebRL, WebAgent, GUI Agents, Mobile Agent

PARADIGM SHIFT

From benchmark-driven evaluation to open-ended exploration—agents actively discovering new knowledge.

Application Overview



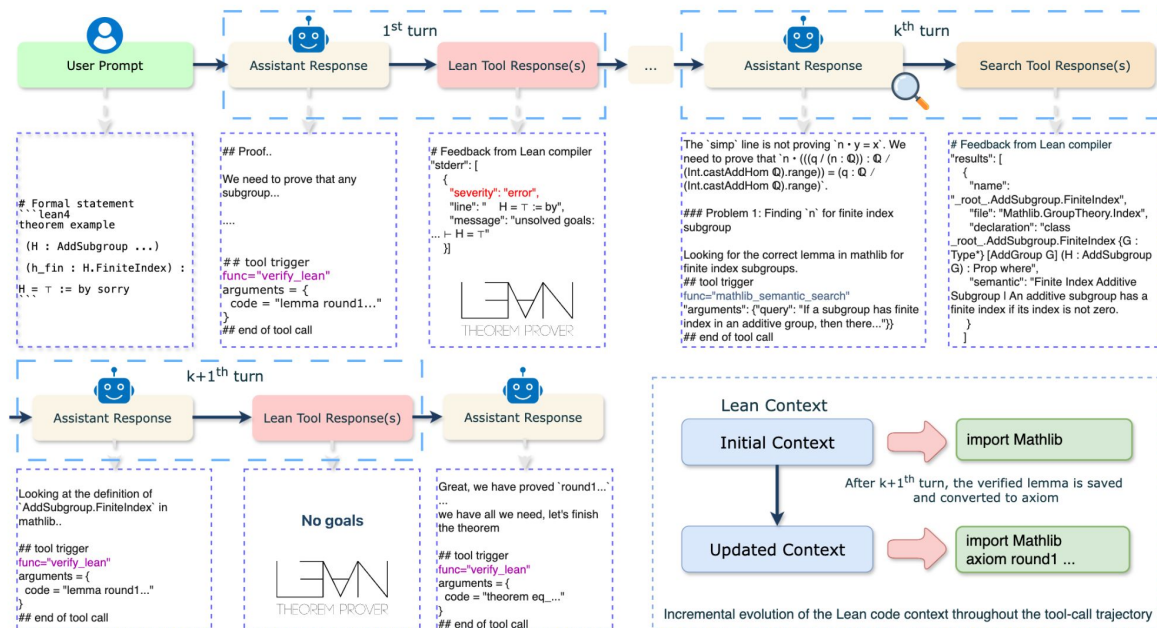
Mathematical

[Ref: *Seed-Prover 1.5: Mastering Undergraduate-Level Theorem*

Proving via Learning from Experience, arXiv 2025]

Agentic capabilities

- code execution for **LEAN** verification and python [foundation layer: tool-use]
- search and retrieval from **MathLib** [foundation layer: tool-use]
- lemma verification [self-evolving layer: feedback]
- verified lemma is stored for re-use [self-evolving layer: memory]
- multi-agent framework: natural language prover, sketch model, agentic prover [collective layer: multi-agent]



Scientific Discovery

Agentic capabilities

- code execution [foundation layer: tool-use]
- search and retrieval for RAG [foundation layer: tool-use]
- self-evolving memory module [self-evolving layer: memory]
- feedback loop [self-evolving layer: feedback]
- multi-agent framework: critic, ideation, analysis, experiment [collective layer: multi-agent]

[Ref: *SciDER: Scientific Data-centric End-to-end Researcher*, arXiv 2026]

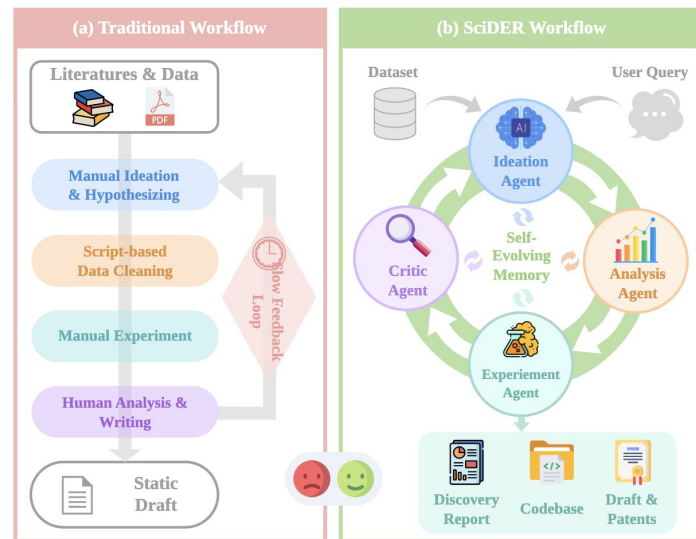


Figure 1: Flowchart of expert-based and LLM-based research lifecycle.

Scientific Discovery

A trending topic: solving very difficult scientific problems, mostly open problems actively studied by human as well
Ex: optimization constants in Mathematics

Agentic capabilities:

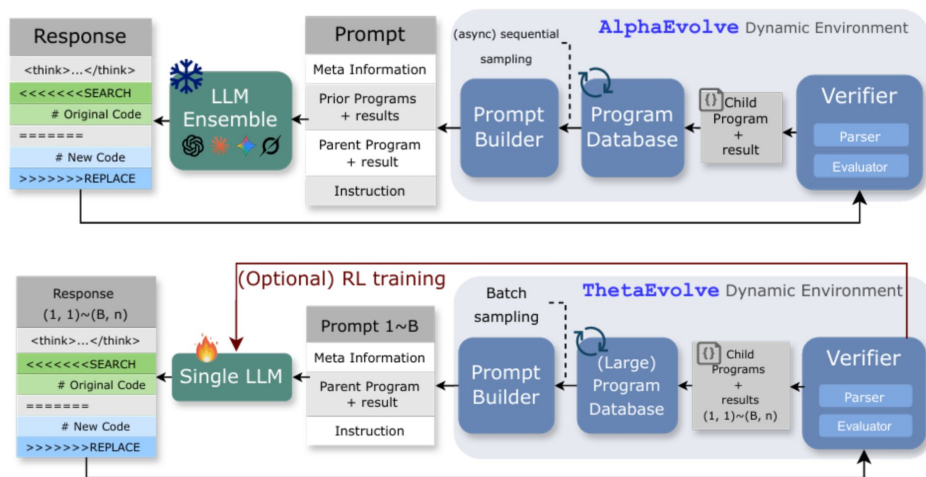
- code execution such as parser and evaluator for verification
- program database serves as memory
- feedback loop that takes verified result and

Additional component:

- test-time training/learning for further improvement

Representative works:

AlphaEvolve, OpenEvolve, ThetaEvolve
TTT-discovery, DeltaEvolve, EvoX



[Ref: AlphaEvolve: A coding agent for scientific and algorithmic discovery, 2025]

[Ref: ThetaEvolve: Test-time Learning on Open Problems, arXiv 2025]

Embodied Agent

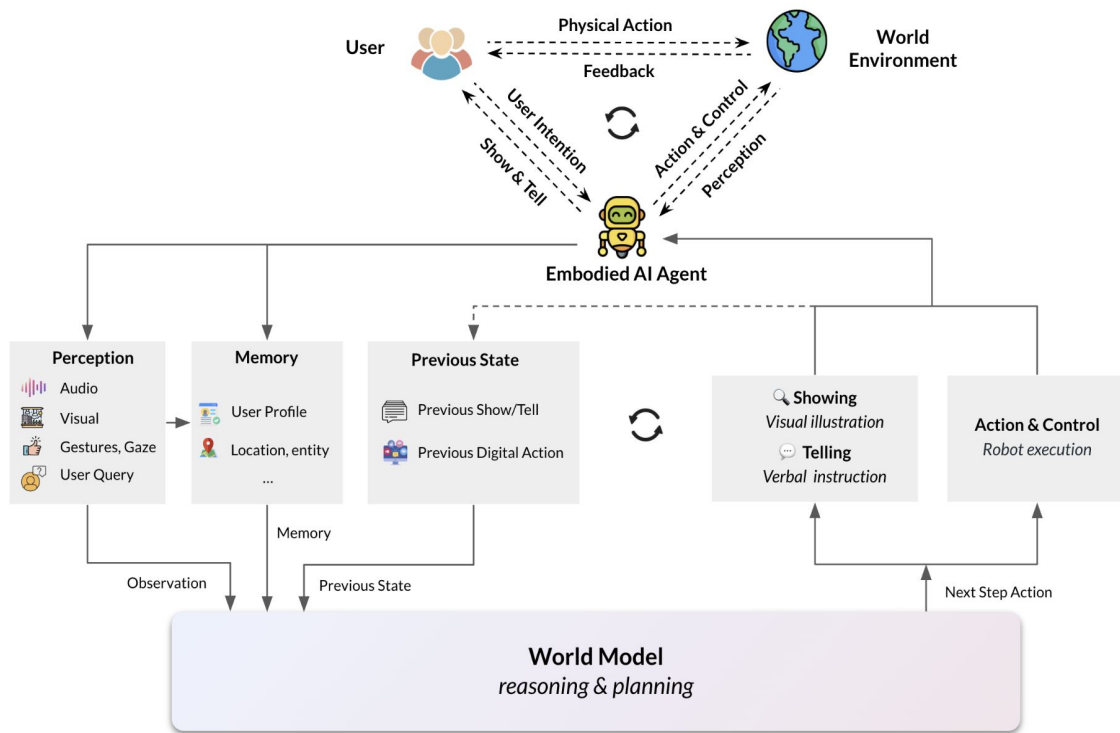
Agentic capabilities

- multi-modal perspection [foundation layer: tool-use]
- interaction with environment and action execution [foundation layer: tool-use]
- personalization entities stored as memory [self-evolving layer: memory]

Integration with world models

- instead of next token prediction, next action prediction
- Visual, Physics and Mental world models
- sees (or senses), predicts, acts and interacts

[Ref: *Embodied AI Agents: Modeling the World*, arXiv 2025]



Autonomous Research

Paradigm shift:

We have seen papers such as
AI Scientist, CycleResearcher, Zochi, AI Scientist v2,
AI Researcher, and DeepScientist
-> producing “academic-format papers”

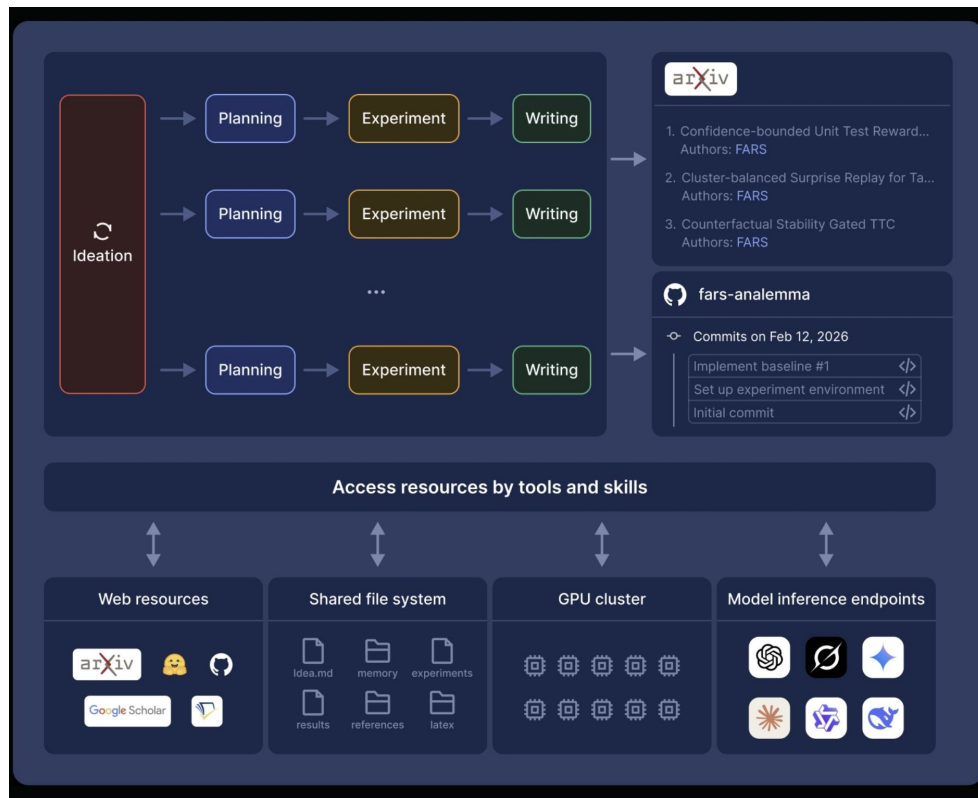
Now, we want not only producing papers that are
“seemingly good” but instead, that can “**efficiently and reliably expand the frontier of knowledge**”.

FARS: Fully Automated Research System, Feb 2026

[Ref: <https://analemma.ai/fars/>]

Some additional agentic capabilities:

- data synthesis [higher-level planning]
- LLM-as-judge [planning]
- web retrieval [tool-use]
- file system as memory backbone [memory]



Benchmarks: Evaluation frameworks capture multi-step reasoning, tool use, and collaborative coordination

⚙️ Mechanism-Centric

TOOL USE

ToolBench, API-Bank, ToolEval

SEARCH & RETRIEVAL

WebArena, FRAMES

MEMORY & LONG-HORIZON PLANNING

LongMemEval, LOCOMO

MULTI-AGENT

MultiAgentBench, BattleAgentBench

Ability-centric Benchmarks (>\$7.1)



Tool Use

- Single-turn
- Multi-turns (>\$7.1.1)



Memory Management

- Long-horizon
- Multi-session (>\$7.1.2)



Multi-agent Collaboration

- Game/Simulation
- Language/Social (>\$7.1.3)

🔗 Application-Level

EMBODIED AGENTS

AgentX, BALROG, AndroidArena, StarDojo

SCIENTIFIC DISCOVERY

DiscoveryWorld, ScienceAgentBench

AUTONOMOUS RESEARCH

DeepSearchQA, HLE, xBench-DeepSearch, ScienceQA, HLE

WEB & OS

WebArena, Webshop, OSWorld, MindAgent

Application-centric Benchmarks (>\$7.2)



Embodied Agents (>\$7.2.1)



Scientific Discovery (>\$7.2.2)



Autonomous Research (>\$7.2.3)



Medical Clinical (>\$7.2.4)



Web Browse (>\$7.2.5)



General Tool-use (>\$7.2.6)



*There is still limitation: Current benchmarks often focus on **final-answer accuracy** rather than **reasoning process quality**, missing the nuance of agentic behavior/often focus on **fixed task distribution**, lacking extensibility to **drifting task environments**.*

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Six frontier challenges define the next decade of agentic reasoning research

01

User-Centric Personalization

Agents must learn individual preferences and communication styles to provide truly personalized assistance over long-term interactions.

02

Long-Horizon Reasoning

Overcoming context limitations to maintain coherent goals and execution over extended timescales (weeks or months).

03

World Model Integration

Integrating learned world models to simulate future states, enabling robust planning and reducing real-world trial-and-error.

04

Multi-Agent Training

Developing new RL frameworks to jointly train multiple agents with shared objectives and emergent communication protocols.

05

Latent Agentic Reasoning

Moving reasoning from explicit token generation to latent space computation for greater efficiency and capability.

06

Governance & Safety

Ensuring alignment, preventing misuse, and establishing accountability frameworks for increasingly autonomous systems.

"The convergence of these directions points toward a future where AI agents serve as genuine collaborative partners."

Conclusion:

A Paradigm Shift in the Second-half of Agentic AI

Four Key Contributions of this Survey:

- **Conceptual Framing:** Formalizing Agentic Reasoning across foundational, self-evolving, and collective layers.
- **Systematic Review:** Analyzing single-agent, adaptive, and multi-agent systems across in-context and post-training dimensions.
- **Applications & Evaluation:** Reviewing real-world applications and benchmarks that illustrate agentic mechanisms.
- **Future Agenda:** Identifying emerging challenges in robustness, trustworthiness, and efficiency for the next generation of agents.

Agentic reasoning marks a fundamental shift from models that passively generate content to autonomous systems that actively perceive, reason, and act. This survey provides a unified roadmap that bridges thought and action, guiding the future development of intelligent agentic systems.

Thank you!
Discussion & QA